European AI Career Intelligence System

Capstone Project for Master in Artificial Intelligence

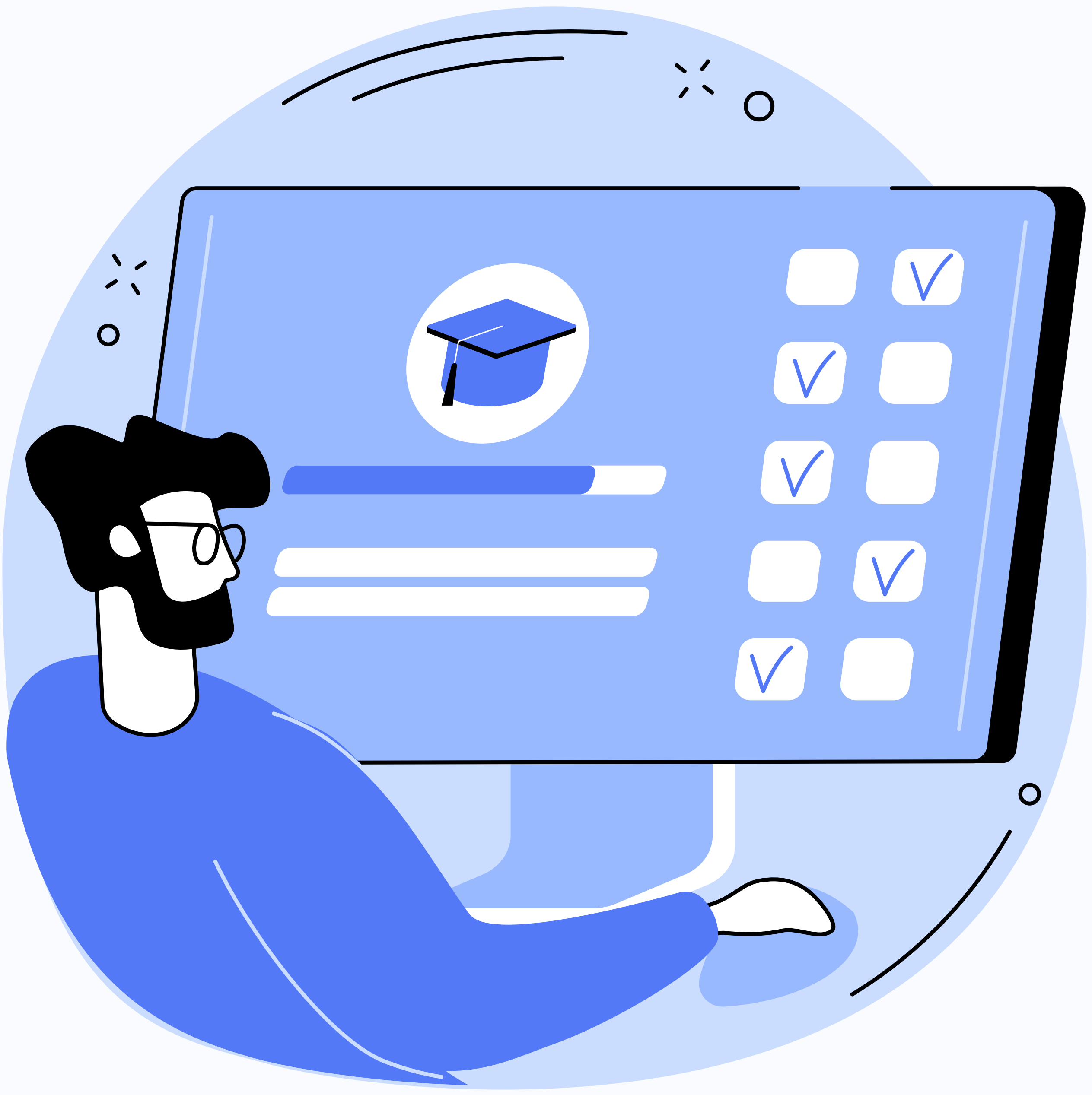
Nina Haide | June 2026

Industry Context

The selected industry is AI talent recruitment and career intelligence in the European market. EU enterprise AI adoption rose from 8% in 2023 to 13.5% in 2024, accelerating demand for qualified AI professionals while the supply of candidates remains constrained. Despite this growth, no candidate-facing career guidance platform exists for Europe. Jobright.ai has proven the model in the United States with 1.25 million users - offering AI-matched jobs, resume tailoring, and a career copilot - but does not operate in Europe. Employer-facing platforms such as Beamery and Eightfold AI target recruiters, not candidates. LinkedIn and Indeed have the richest data but prohibit programmatic access entirely. Adzuna, a British job aggregator operating in 19 countries and serving as an official UK ONS data partner, is the only viable open European data source - and the foundation of this project.

Problem Definition

The core problem is a three-sided information asymmetry. Candidates do not know which AI roles their skills qualify them for - a finance analyst with SQL and Python may be a natural fit for a Data Analytics role but will never find it through a keyword job search. Employers cannot define roles consistently - job titles are unreliable proxies for actual skill demand, leading to mismatched applications and wasted screening time. And the market has no structured, data-driven taxonomy of AI roles specific to the European context - existing frameworks are US-derived and too generic to reflect local hiring norms. This problem is well-suited to AI because the signal is latent: the meaningful structure of the job market is not explicitly labelled anywhere and must be learned from data at a scale that exceeds human capacity to analyse manually.



Integrated AI System Overview

EACIS is a six-layer AI pipeline that takes a candidate's resume or skill description as input and produces a structured career report - including an archetype match, personalised recommendation, semantically similar real job postings, current live openings, and salary context - all orchestrated by a ReAct agent without manual intervention between steps.

01 Data Infrastructure Layer

The Adzuna API collects 1,328 live job postings across six European countries and enriches truncated descriptions from source pages, increasing average description length from 500 to 3,400 characters.

02 Market Intelligence Layer

Real salary data is loaded from the ai-jobs.net DACH dataset, converts all values to EUR using live ECB exchange rates, and segments salary ranges by archetype informed by the ANOVA finding that remote work significantly affects compensation.

03 Skill Taxonomy and Archetype Layer

A 50-skill taxonomy is applied to extract skills from job descriptions and uses k-Means clustering to discover five role archetypes from the data - not from a predefined list.

Integrated AI System Overview

04 Semantic Representation Layer

The semantic representation layer (Project 4) encodes every job description as a 768-dimensional DistilBERT vector and indexes all 1,088 vectors in a FAISS index for fast similarity search at query time.

05 Recommendation Logic Layer

The recommendation logic layer (Project 5) routes candidates into one of three fit levels - strong, partial, or none - based on cosine similarity to archetype centroids, and generates structured career guidance conditioned on that fit level and retrieved context.

06 Agentic Orchestration Layer

The agentic orchestration layer (Project 6) is a ReAct agent that runs six tools in sequence - extract skills, match archetype, retrieve similar postings, generate recommendation, fetch live jobs, look up salary - maintaining a state dictionary across all steps and logging a full reasoning trace in every output.

The offline pipeline runs once to build system artefacts: job postings are collected, skills extracted, archetypes clustered, descriptions encoded, the FAISS index built and salary ranges pre-computed. The online pipeline runs per candidate query in approximately 30 seconds, with each tool's output feeding into the next step's reasoning. The reasoning trace - the agent's full Thought-Action-Observation log - is included in every career report, making every decision visible and auditable.

Integration of Prior Capstones

EACIS integrates all six prior capstone projects, and the integration is compositional rather than additive - removing any single layer breaks the system's ability to produce its output.

Project 1 established that Adzuna truncates descriptions at 500 characters - a data quality finding that shaped every downstream architectural decision.

Project 2 proved statistically that remote work mode significantly affects salary ($F=4.56$, $p=0.011$), giving the system a real analytical basis for salary segmentation rather than estimates.

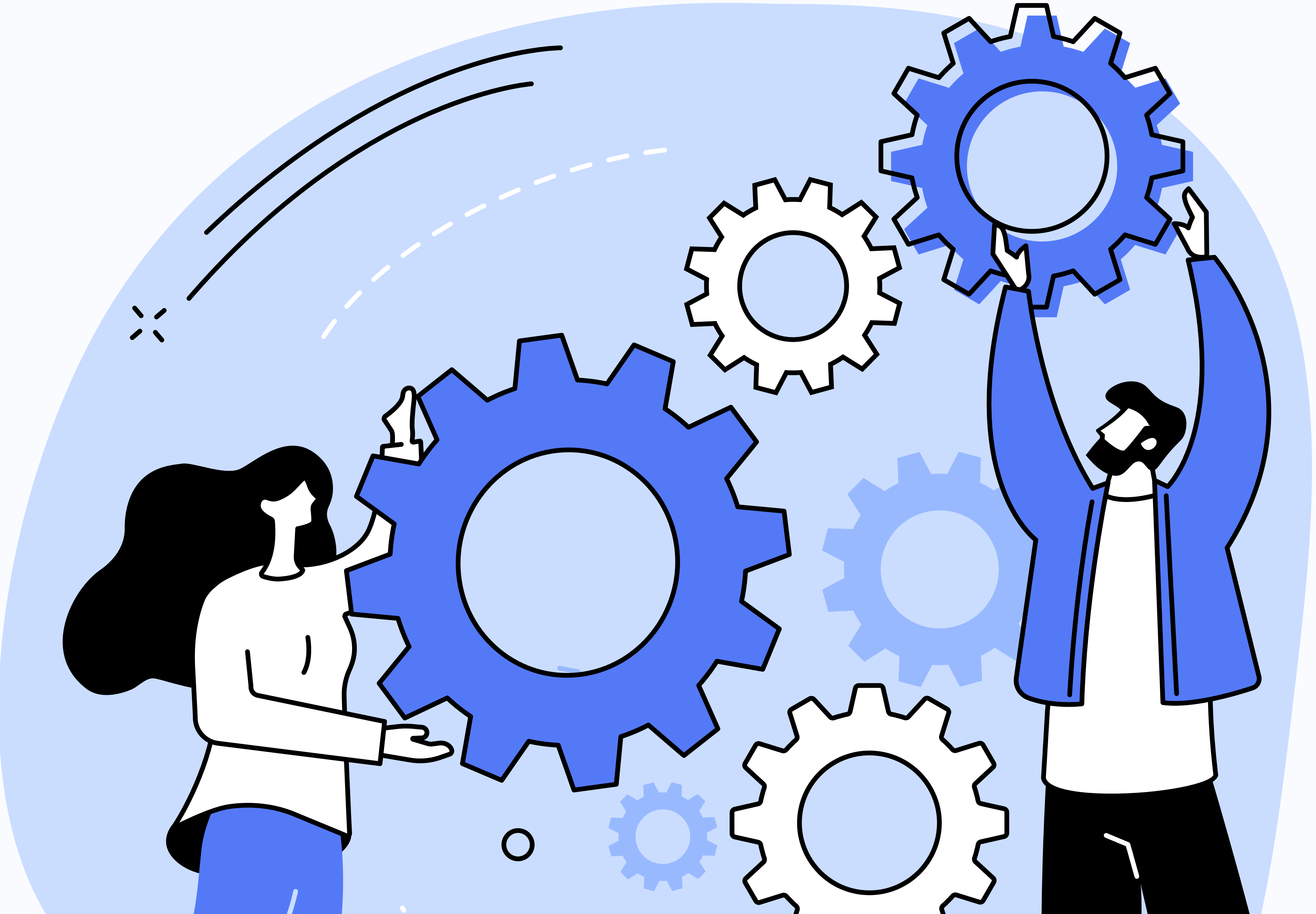
Project 3 showed empirically that AI consulting and PM roles share skill profiles with engineering roles - validating the core hypothesis that job titles are unreliable and skill-based matching is necessary.

Project 4 demonstrated through direct comparison that DistilBERT produces more coherent clusters than TF-IDF, justifying the Transformer choice in production.

Project 5 proved through failure that small language models hallucinate in career guidance contexts, which led to the template approach - a more reliable and honest design decision.

Project 6 synthesised all prior components into the ReAct agent, making the system end-to-end autonomous.

Key Technical Decisions and Tradeoffs



DistilBERT over TF-IDF - Project 4 compared both directly. DistilBERT produced more coherent clusters. The tradeoff is compute cost: 10–20 minutes to build the FAISS index on CPU versus seconds for TF-IDF. Accepted because semantic quality matters more than build time for a one-off offline step.

Templates over GPT-2 - GPT-2 was tested and produced hallucinations. For a career guidance tool where a wrong recommendation has real consequences, reliability matters more than generative variety. A harder decision than it sounds - it means every strong-fit candidate receives structurally identical output.

English-only DistilBERT over mBERT - 85%+ of collected postings are in English even for DACH markets. Multilingual models are 3× larger and impractical on CPU. Documented as a known limitation rather than a hidden one.

Single-agent ReAct over multi-agent - Sequential execution keeps the reasoning trace legible and the state dictionary simple. The tradeoff is latency - tools cannot run in parallel. Acceptable for a career orientation tool where session time is not critical.

Keyword taxonomy over NER - The 50-skill taxonomy is interpretable and debuggable. A trained NER model would have higher coverage but lower transparency. Given the EU AI Act requirement for explainability in recruitment AI, interpretability was prioritised. The coverage gap is the biggest known limitation.

Ethical Considerations and Responsible AI

Recruiter misuse is the primary risk. Similarity scores designed to guide candidates could be repurposed to rank or filter applicants - encoding the biases in 80% UK and German training data into hiring decisions. The mitigation is architectural: no recruiter-facing interface exists, and the responsible use boundary is documented across all project artefacts.

Overconfident output is the second risk. The system says "your profile aligns strongly with..." based on a 50-keyword match with no uncertainty quantification. The mitigation is the full reasoning trace included in every report - candidates see exactly what was detected and why - and explicit framing as a first-pass orientation tool, not a definitive career assessment.

EU AI Act classifies recruitment AI as high-risk under Annex III, with compliance enforceable from August 2026. EACIS was designed to meet these requirements from the outset - session isolation, explainable reasoning traces and no recruiter interface eliminate the highest-risk deployment scenarios before they can occur.

Evaluation, Limitations and Risks

The system was evaluated across four candidate profiles. All seven integration layers were exercised across all profiles, confirming end-to-end functionality.

Strength: The core hypothesis was validated. An automotive engineer with only Python and computer vision - no formal AI background - correctly matched to Deep Learning and AI Research with a strong fit score and relevant live jobs returned. Skill-based matching surfaces pathways that keyword search cannot.

Key limitation: The 50-skill taxonomy underestimates non-standard profiles. The same automotive engineer was reported at 25% archetype coverage despite genuine high competency - terms like embedded systems and C++ are simply absent from the taxonomy. This propagates through every downstream component.

Production risks: 80% of training data is from the UK and Germany, meaning archetypes primarily reflect those markets. The English-only DistilBERT model partially mis-tokenises German and French postings. Both require a multilingual model and broader data collection before any production deployment.

Professional Relevance and next Steps

This project demonstrates end-to-end AI system ownership across six domains - data engineering, statistical analysis, machine learning, deep learning, generative AI, and agentic orchestration - integrated into a single defensible product. Every architectural decision is justified by experimental evidence from a prior project, not by preference. The GPT-2 failure is documented honestly rather than hidden. EU AI Act compliance was designed in from the start. These are the practices that distinguish reliable AI practitioners from those who overstate system capabilities.

Next Steps

The most impactful improvement would be replacing the fixed skill list with a model that can actually read and understand resume text - this would fix the case where an automotive engineer with strong relevant skills scores only 25% coverage because their terminology doesn't match our keywords. Job search results should also be personalised to the candidate's actual skills rather than returning the same postings for everyone in the same archetype. For non-English candidates, the current model needs to be swapped for one that handles German and French properly. The output should show how well the candidate matched all five archetypes, not just the best one, so borderline cases are more transparent. Finally, the web application is about to be deployed - that would make the system publicly accessible rather than notebook-only.

Conclusion

EACIS started as a question - why does no European equivalent of Jobright.ai exist - and became a six-project integrated pipeline that provides a working answer. Every component was tested before it was trusted and every failure produced a better design decision than the original plan.

The European AI job market has a genuine, solvable information asymmetry problem.

The tools to address it exist. What remains is scale, multilingual coverage and responsible deployment.

Thank you!

Presentation By Nina Haide